EVENT MANAGEMENT SYSTEM

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Colleges, universities, and organizations conduct various events such as seminars, workshops, hackathons, conferences, technical fests, and cultural programs. Managing these events manually can result in scheduling conflicts, duplicate registrations, communication gaps, and inefficient data management. The proposed Event Management System (EMS) is a web-based platform designed to automate event creation, searching, registration, scheduling, resource allocation, and reporting by integrating advanced algorithmic techniques. The system uses string algorithms such as KMP and Rabin-Karp for efficient event and participant searching. Dynamic Programming, particularly Edit Distance, is used for fuzzy search and correcting user queries. Suffix Arrays and LCP can support efficient similarity analysis of event descriptions and documents. Network Flow algorithms such as Dinic's algorithm are applied to resource allocation, participant assignment, and conflict-free scheduling. For complex scheduling and allocation problems, NP-hard concepts and approximation algorithms such as the Vertex Cover 2-approximation provide practical solutions when exact solutions are computationally expensive. Randomised algorithms such as Miller-Rabin primality testing and randomized hashing demonstrate efficient probabilistic processing, while parallel algorithms such as parallel reduction and prefix-sum can improve the processing of large event datasets. By combining these techniques, the EMS provides efficient searching, scheduling, allocation, data processing, and management, reducing administrative workload and providing a reliable and user-friendly experience for organizers and participants.

Signature of the Guide